Genomic Characterization and Survival Analysis of Lung Adenocarcinoma Using TCGA Data

Introduction

Lung adenocarcinoma (LUAD) represents the most common histological subtype of lung cancer, accounting for approximately 40% of all cases globally (Travis et al., 2015). Despite advances in targeted therapies and immunotherapies, the 5-year survival rate remains poor, highlighting the need for better understanding of the molecular drivers and prognostic factors of this disease (Siegel et al., 2023). The Cancer Genome Atlas (TCGA) provides a valuable resource for comprehensive molecular profiling of cancers, including LUAD (Cancer Genome Atlas Research Network, 2014). This study aims to characterize the genomic landscape of LUAD and identify genetic alterations associated with patient outcomes (Skoulidis and Heymach, 2019).

Methods

Data Source

Data were obtained from the TCGA-LUAD cohort via cBioPortal, including clinical information, somatic mutations, and gene expression data for 585 patients. Analysis focused on key driver genes previously implicated in lung adenocarcinoma: EGFR, KRAS, TP53, STK11, and KEAP1.

Clinical Analysis

Patient demographics and clinical characteristics were analyzed to establish baseline cohort profiles. Distributions of age, gender, smoking history, race, ethnicity, and pathologic stage were visualized using histograms, pie charts, and bar plots.

Mutation Analysis

Somatic mutation data were analyzed using the maftools package in R. Mutation frequencies were calculated for each key gene, and mutation patterns were visualized using lollipop plots to identify distribution across functional protein domains. Co-occurrence patterns between mutations were examined to identify potential interactions.

Survival Analysis

Kaplan-Meier analyses were performed to assess relationships between clinical factors (stage, gender, smoking status) and genomic alterations (mutations in key genes) with overall survival. Log-rank tests were used to determine statistical significance ($p < 0.05$). Multivariate Cox proportional hazards models were employed to identify independent prognostic factors.

Gene Expression Analysis

RNA-seq data (FPKM values) were analyzed to identify genes with variable expression across the cohort. Principal component analysis and hierarchical clustering were performed to explore expression patterns, and the top variable genes were characterized.

Results

Patient Characteristics

The cohort consisted of 585 LUAD patients with a balanced gender distribution (274 females, 235 males) and median age of 67 years. The majority of patients were White (n=393), with fewer Black or African American (n=53), Asian (n=8), and American Indian or Alaska Native (n=1) patients. Most patients were non-Hispanic/Latino (n=389). Smoking history was documented for most patients, with a median of 33 pack-years (Imielinski et al., 2012).

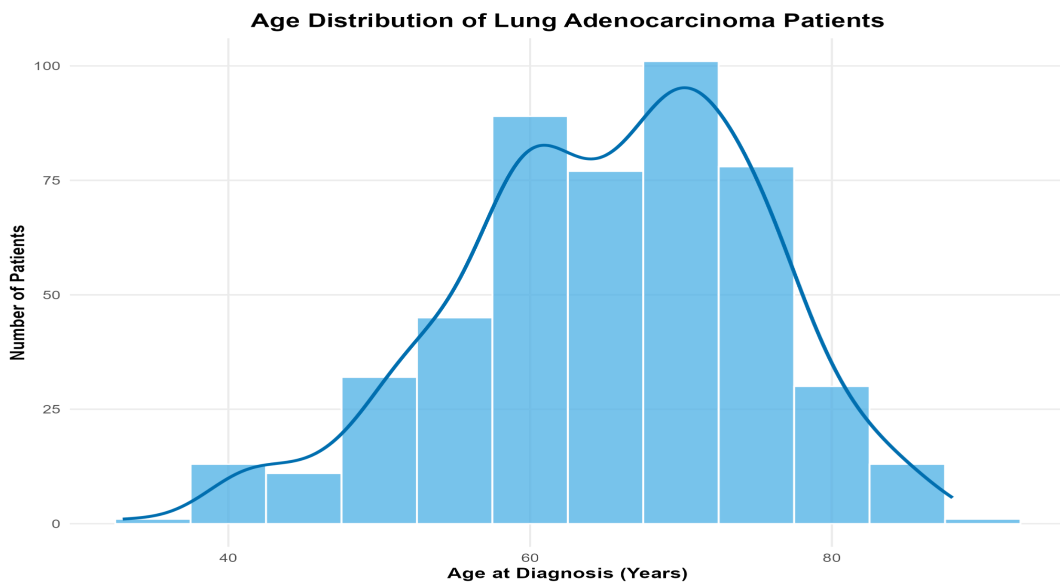


Figure 1. Age distribution of lung adenocarcinoma patients from the TCGA cohort. Histogram shows patient ages at diagnosis, with a median age of 67 years. The distribution is approximately normal, with most patients diagnosed between 60-75 years of age, reflecting the typical late-onset pattern of lung cancer. This age distribution aligns with the known epidemiology of lung adenocarcinoma, which primarily affects older adults (n=585).

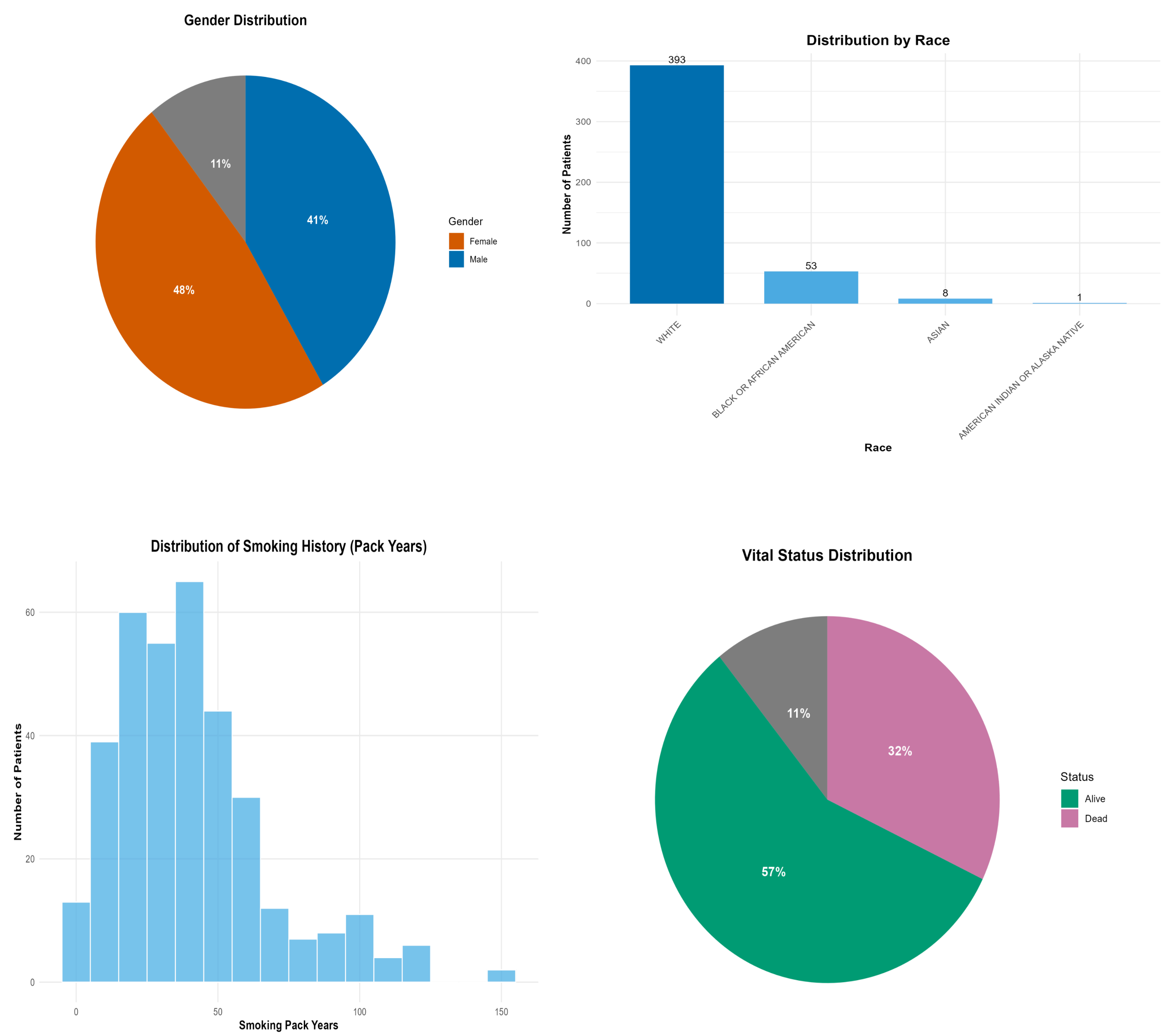
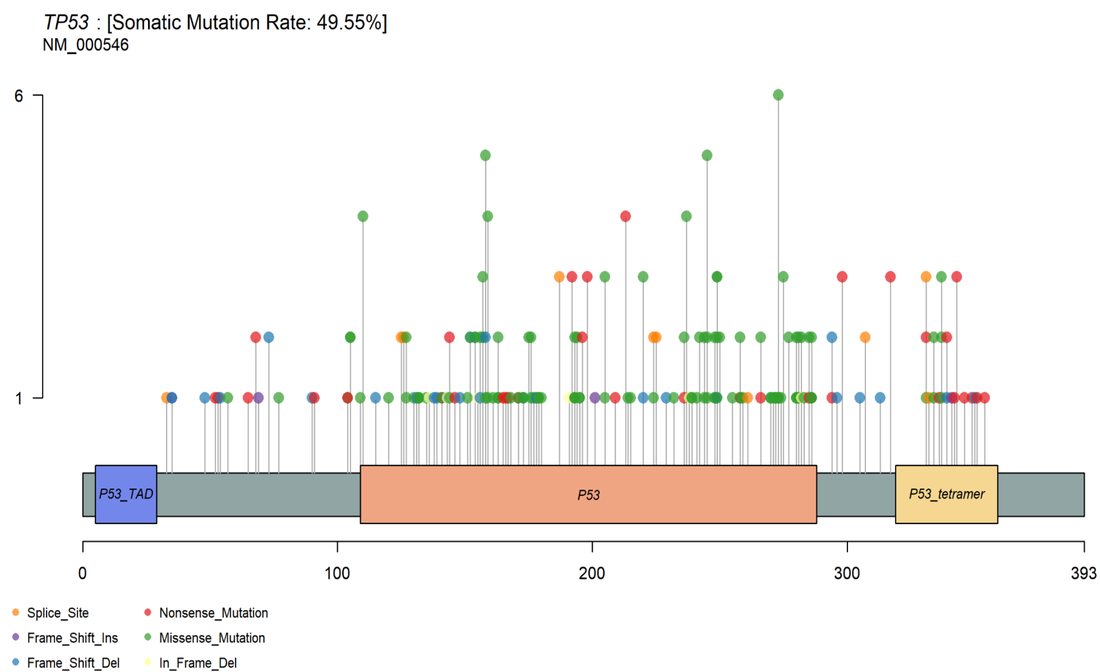
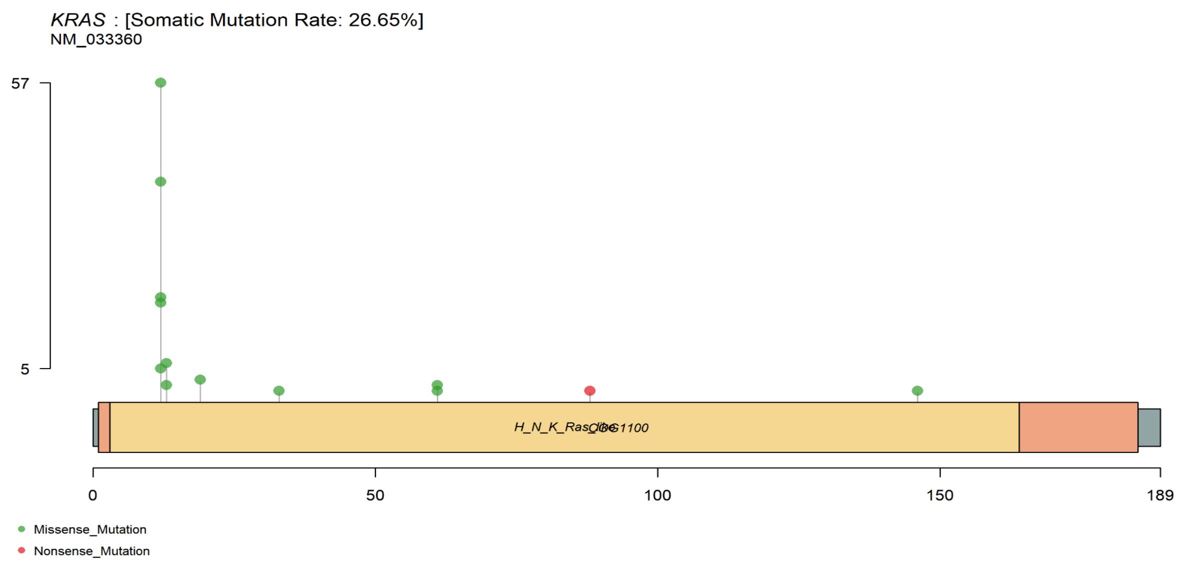


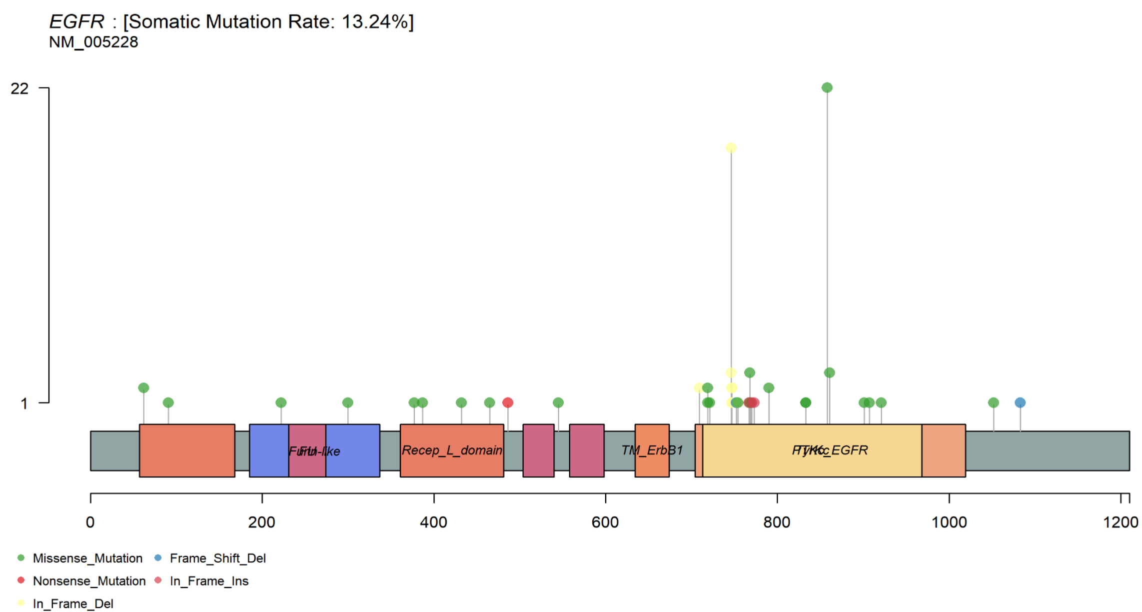
Figure 2. Demographic characteristics of lung adenocarcinoma patients. A) Gender distribution shows a slight female predominance (48% female, 41% male, 11% data not available). B) Racial distribution indicates the majority of patients were White (n=393), followed by Black or African American (n=53), Asian (n=8), and American Indian/Alaska Native (n=1). C) Smoking history distribution (pack-years) reveals most patients had significant smoking exposure, with the highest frequency between 20-50 pack-years. D) Vital status shows 57% of patients were alive and 32% deceased at last follow-up, with 11% unknown status.

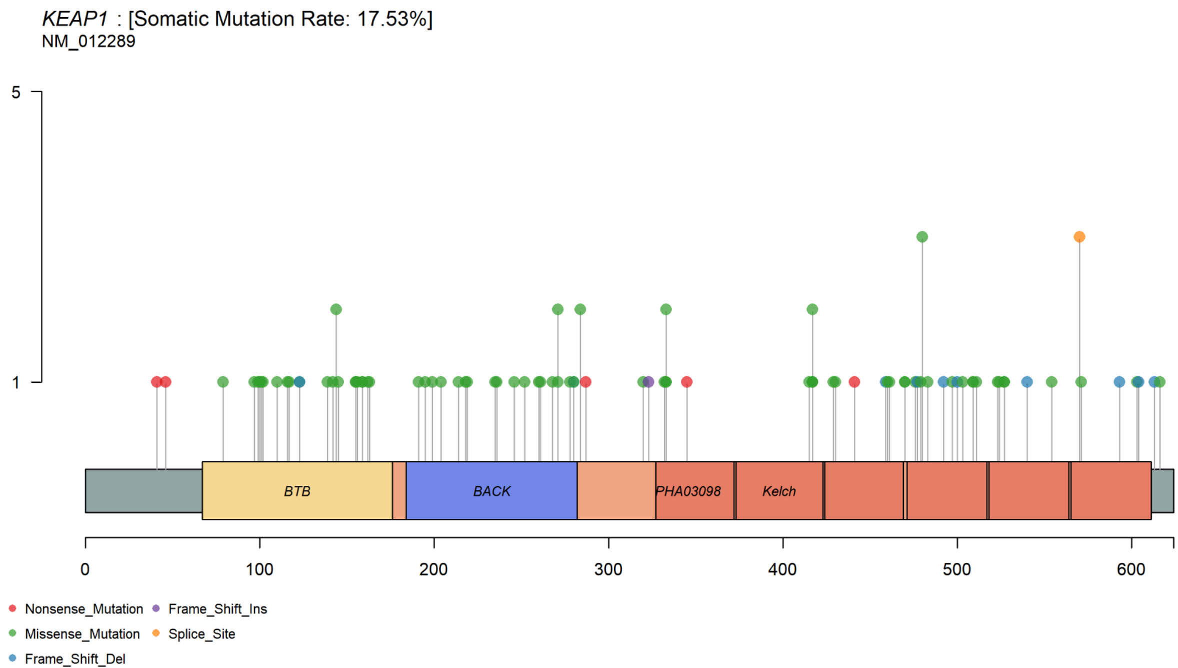
Mutation Landscape

Analysis revealed TP53 as the most frequently mutated gene (50%), followed by KRAS (27%), KEAP1 (17%), EGFR (13%), and STK11 (12%). Lollipop plots demonstrated that TP53 mutations were predominantly located in the DNA-binding domain, KRAS mutations clustered at codons 12-13 in the GTPase domain, EGFR mutations concentrated in the tyrosine kinase domain, KEAP1 mutations were distributed throughout the protein, and STK11 mutations showed no clear hotspot pattern (Jamal-Hanjani et al., 2017).




KRAS : [Somatic Mutation Rate: 26.65%]
NM_033360
57
5
H_N_K_Ras_like
0
50
100
150
189
Missense_Mutation
Nonsense_Mutation


EGFR : [Somatic Mutation Rate: 13.24%]
NM_005228
22
1
Recep_L_domain
TM_ErbB1
0
200
400
600
800
1000
1200
Missense_Mutation
Frame_Shift_Del
Nonsense_Mutation
In_Frame_Ins
In_Frame_Del


KEAP1 : [Somatic Mutation Rate: 17.53%]
NM_012289
5
1
BTB
BACK
PHA03098
Kelch
0
100
200
300
400
500
600
Nonsense_Mutation
Frame_Shift_Ins
Missense_Mutation
Splice_Site
Frame_Shift_Del

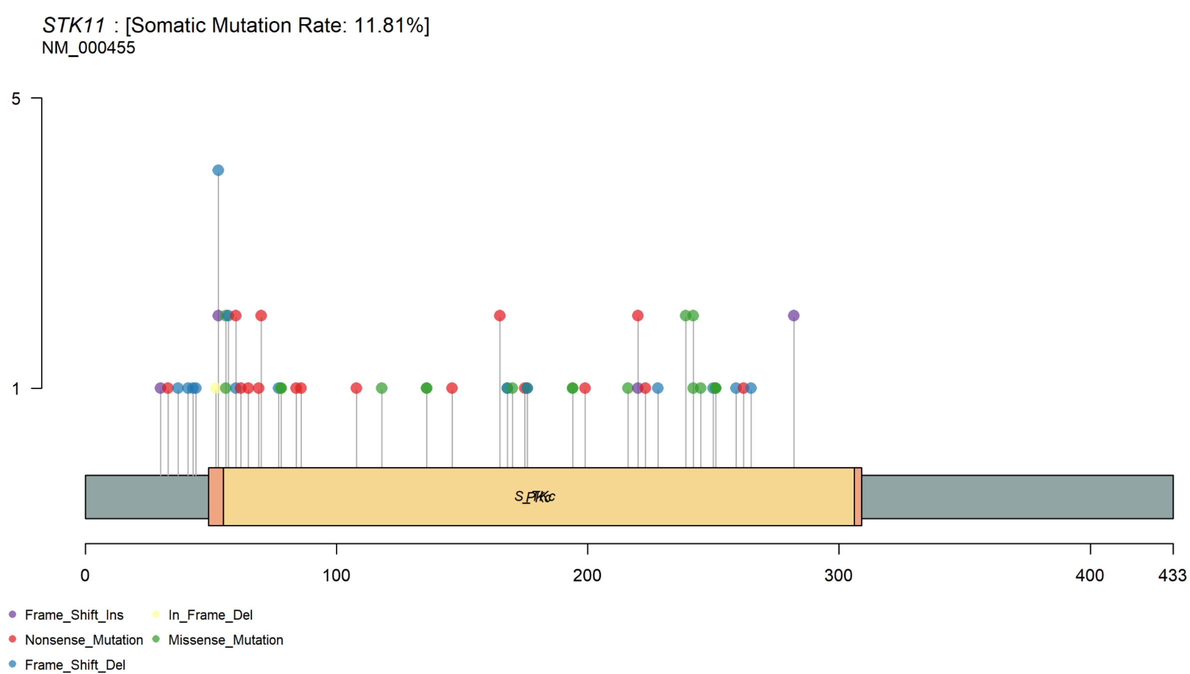
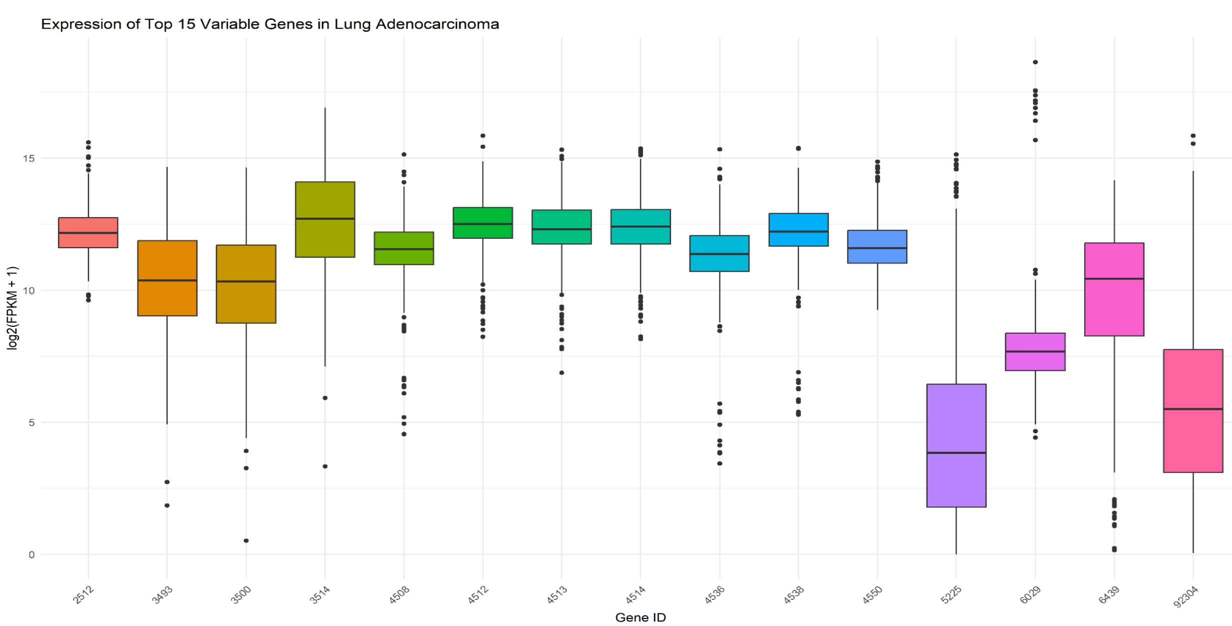

STK11 : [Somatic Mutation Rate: 11.81%]
NM_000455
5
1
0
100
200
300
400
433
Frame_Shift_Ins
In_Frame_Del
Nonsense_Mutation
Missense_Mutation
Frame_Shift_Del

Figure 3. Lollipop plots illustrating mutation patterns in key driver genes. These plots show the distribution and frequency of mutations across protein domains for each gene. A) TP53 (mutation rate: 49.55%) shows concentration of mutations in the DNA-binding domain, consistent with its role as a tumor suppressor. B) KRAS (mutation rate: 26.65%) exhibits hotspot mutations primarily at codons 12 and 13, which are known to constitutively activate KRAS signaling. C) EGFR (mutation rate: 13.24%) displays mutations primarily in the tyrosine kinase domain, which can lead to ligand-independent receptor activation. D) KEAP1 (mutation rate: 17.53%) and E) STK11 (mutation rate: 11.81%) show more distributed mutation patterns across their respective protein domains.

Gene Expression Analysis

Expression analysis identified the top variable genes across the cohort. Hierarchical clustering of these genes revealed distinct expression patterns that may represent different molecular subtypes of LUAD (Wilkerson et al., 2012).



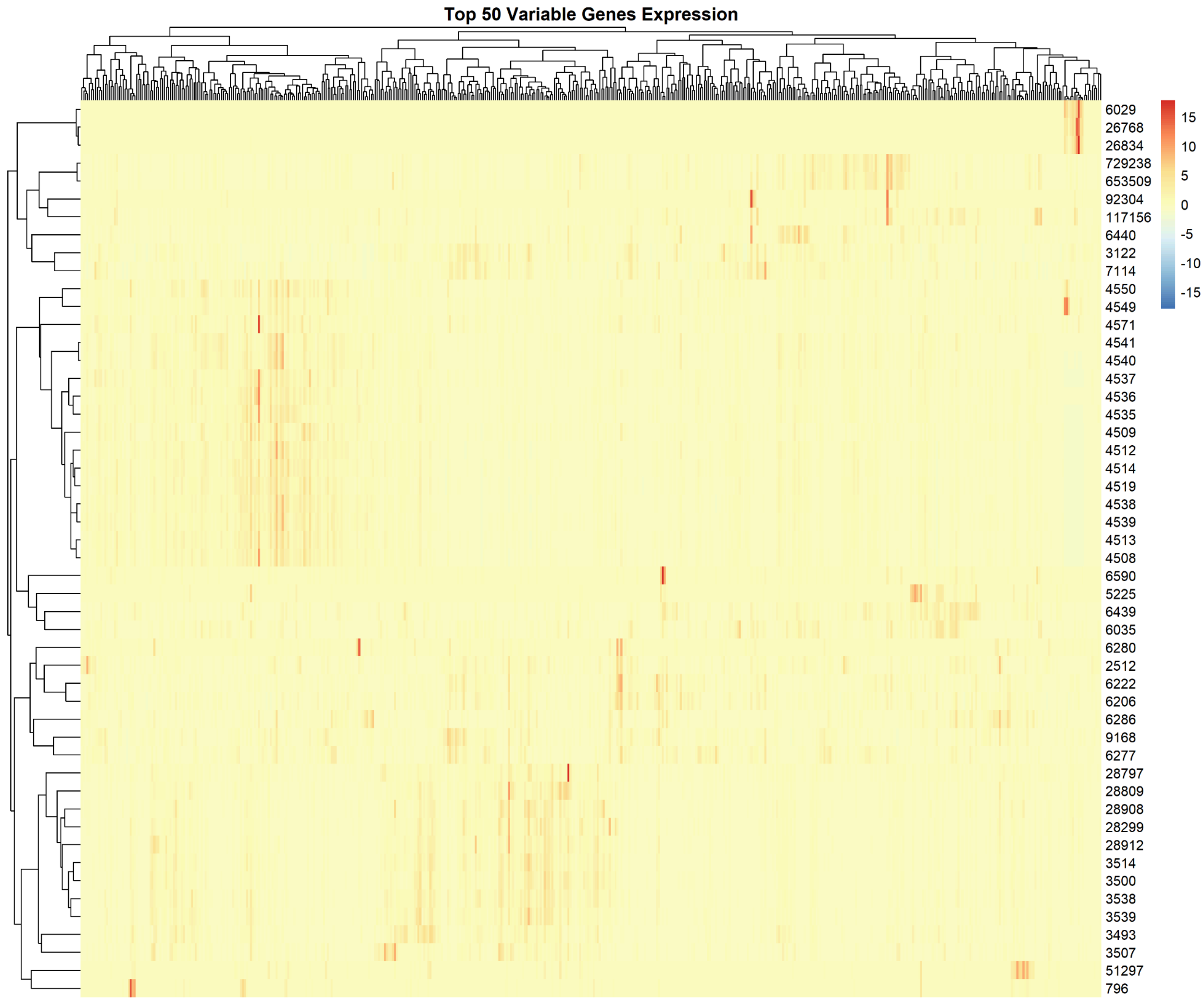
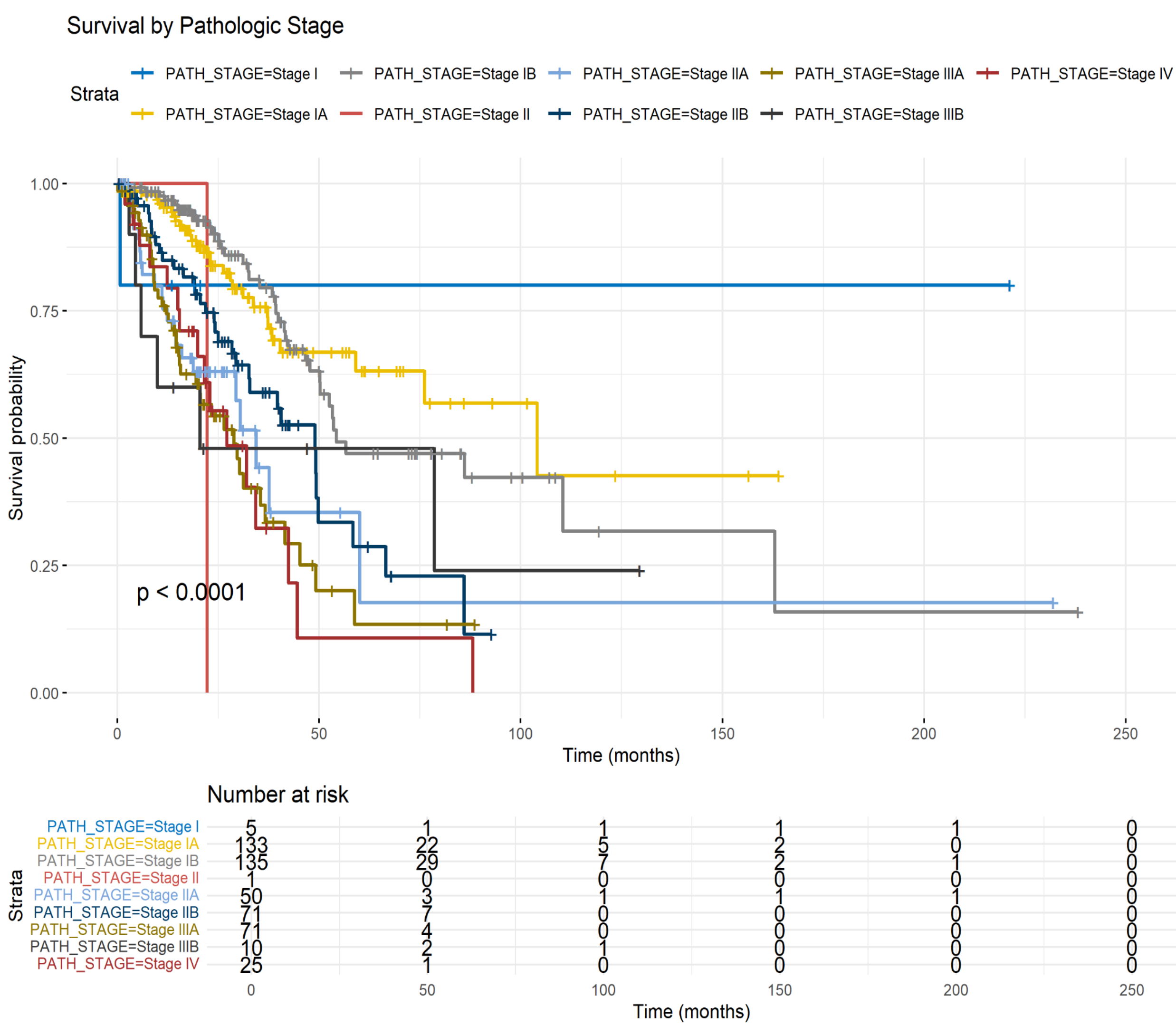
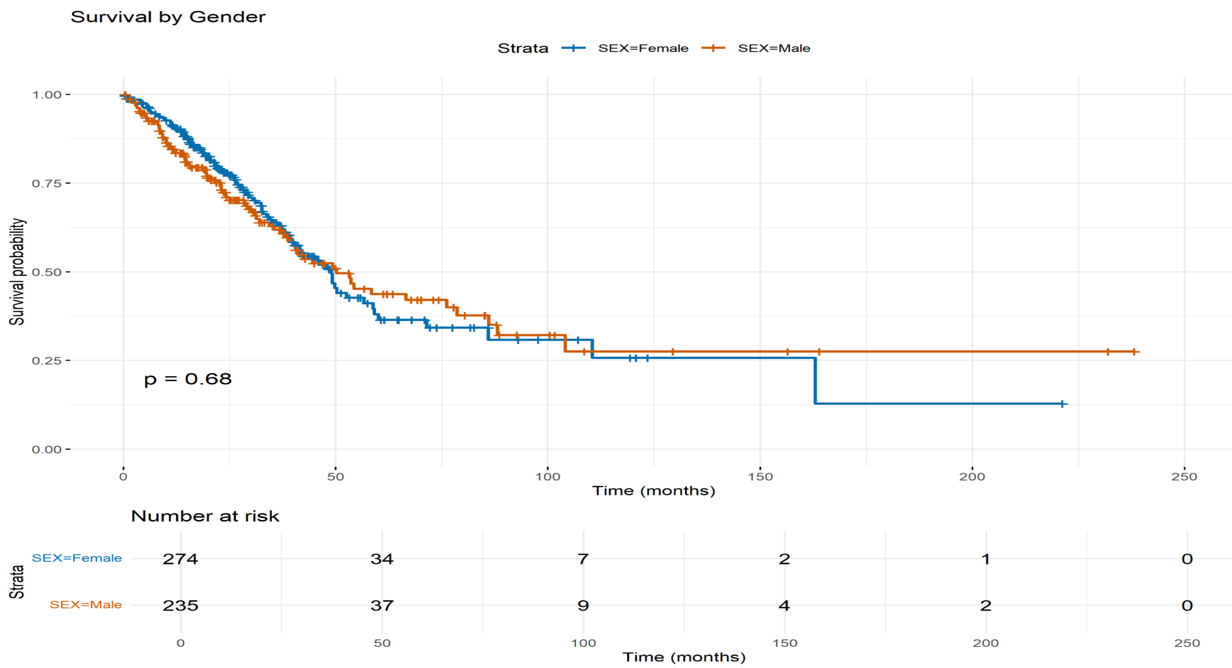


Figure 4. Expression patterns of the top variable genes in lung adenocarcinoma. A) Boxplot showing the expression levels (log2(FPKM+1)) of the top 15 variable genes across all samples. Each gene exhibits distinct expression patterns and variability, reflecting the heterogeneity of lung adenocarcinoma. B) Heatmap of the top 50 variable genes across all samples. Hierarchical clustering of both genes and samples reveals distinct expression patterns that may correspond to different molecular subtypes of lung adenocarcinoma. Red indicates high expression and blue indicates low expression, with each row representing a gene and each column representing a patient sample.

Survival Analysis

Survival analysis by clinical factors revealed that the pathologic stage was significantly associated with outcome ($p < 0.0001$), with earlier stages showing better survival. Gender and smoking status did not significantly impact survival ($p = 0.68$ and $p = 0.84$, respectively). Among the analyzed genes, only TP53 mutations showed a statistically significant association with worse survival ($p = 0.0393$), while EGFR ($p = 0.2772$), KRAS ($p = 0.7073$), STK11 ($p = 0.1815$), and KEAP1 ($p = 0.3129$) mutations did not significantly impact survival in univariate analysis (Gu et al., 2018).




Survival by Gender
Strata SEX=Female SEX=Male
Survival probability
1.00
0.75
0.50
0.25
0.00
p = 0.68
0 50 100 150 200 250
Time (months)
Number at risk
Strata
SEX=Female 274 34 7 2 1 0
SEX=Male 235 37 9 4 2 0
0 50 100 150 200 250
Time (months)

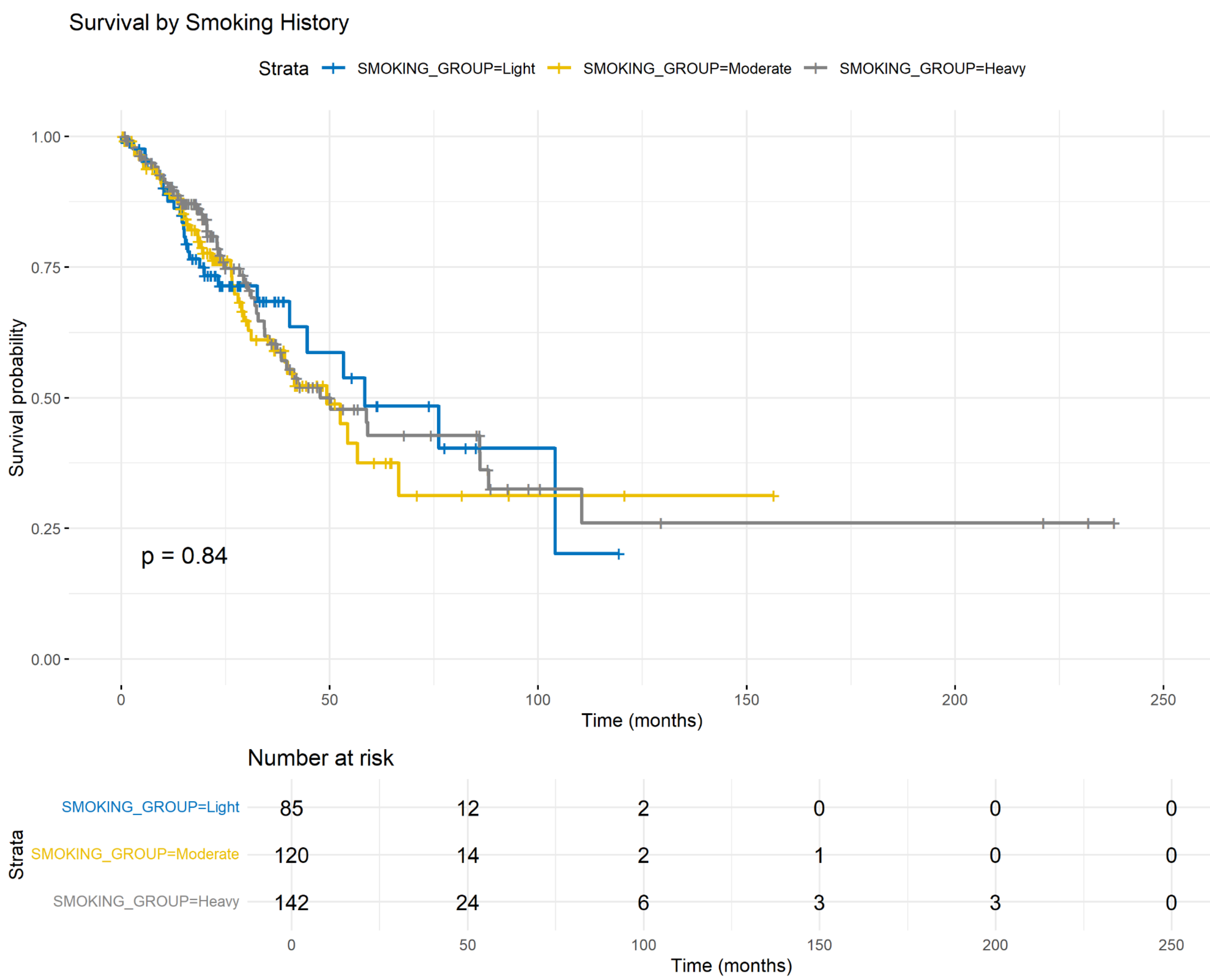

Survival by Smoking History
Strata SMOKING_GROUP=Light SMOKING_GROUP=Moderate SMOKING_GROUP=Heavy
Survival probability
1.00
0.75
0.50
0.25
0.00
p = 0.84
0 50 100 150 200 250
Time (months)
Number at risk
Strata
SMOKING_GROUP=Light 85 12 2 0 0 0
SMOKING_GROUP=Moderate 120 14 2 1 0 0
SMOKING_GROUP=Heavy 142 24 6 3 3 0
0 50 100 150 200 250
Time (months)

Figure 5. Survival analysis in lung adenocarcinoma patients. Kaplan-Meier curves showing overall survival stratified by A) pathologic stage, which was strongly associated with outcome ($p < 0.0001$), with earlier stages showing better survival; B) gender, which showed no significant difference in survival between males and females ($p = 0.68$); C) smoking history, which did not significantly impact survival ($p = 0.84$); and D) TP53 mutation status, where patients with mutated TP53 had significantly worse survival outcomes compared to those with wild-type TP53 ($p = 0.0393$).

Multivariate Analysis

Cox proportional hazards modeling identified pathologic stage (Stage II: HR = 2.15, 95% CI: 1.35-3.41, $p = 0.001$; Stage III: HR = 3.02, 95% CI: 1.87-4.87, $p < 0.0001$; Stage IV: HR = 2.69, 95% CI: 1.31-5.50, $p = 0.007$) as the strongest independent prognostic factor. TP53 mutation (HR = 1.43, 95% CI: 0.97-2.12, $p = 0.072$) showed a trend toward worse outcomes but did not reach statistical significance in the multivariate model. EGFR, KRAS, STK11, and KEAP1 mutations, as well as gender and smoking status, were not significant independent prognostic factors (Liu et al., 2018).

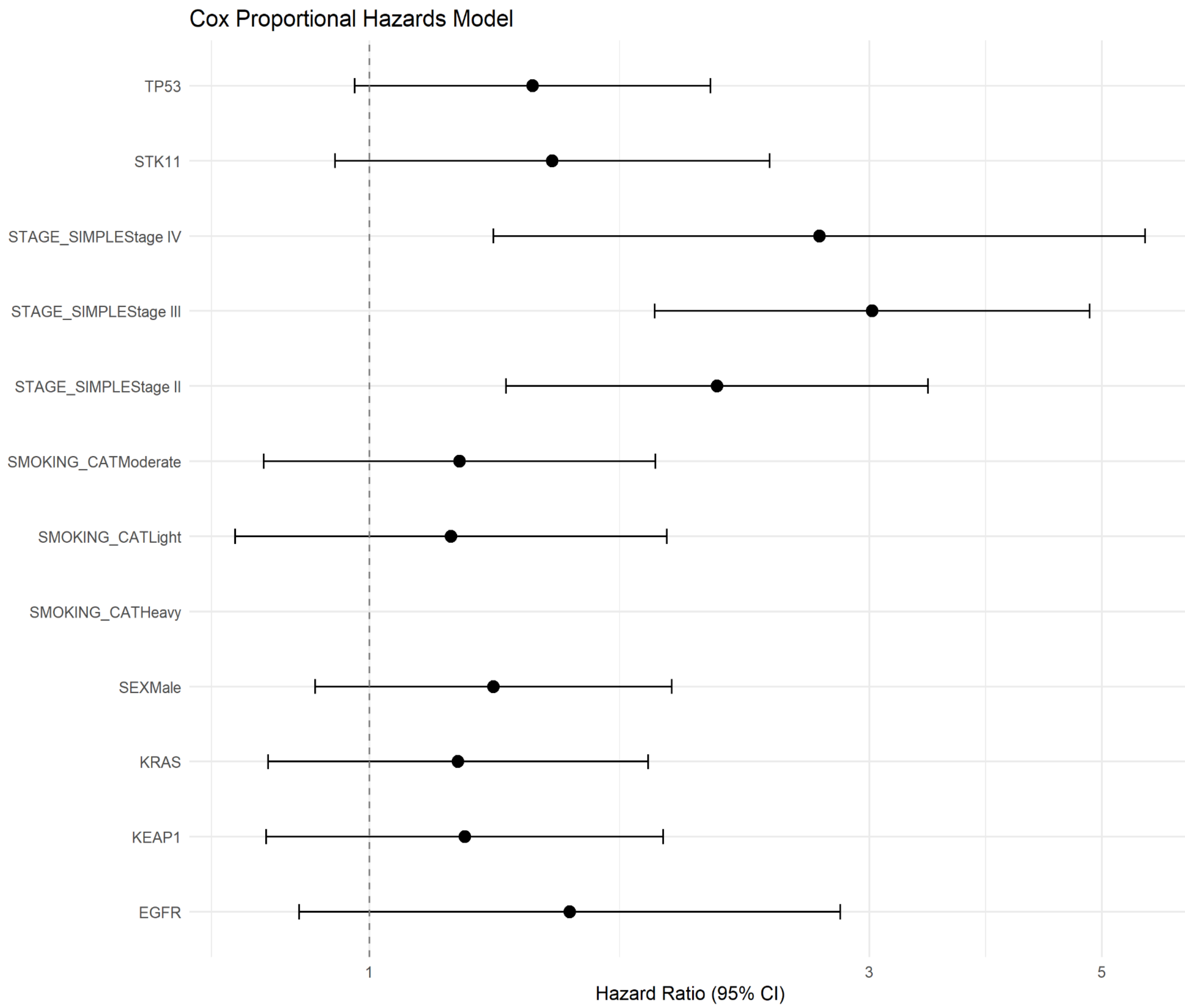


Figure 6. Forest plot of Cox proportional hazards model for lung adenocarcinoma. Multivariate analysis showing hazard ratios with 95% confidence intervals for key clinical and genomic factors. Pathologic stage emerged as the strongest independent prognostic factor, with Stage III showing the highest hazard ratio (HR = 3.02). TP53 mutation showed a trend toward worse outcomes (HR = 1.43) but did not reach statistical significance (p = 0.072) in this multivariate model when adjusting for other factors.

Discussion

This comprehensive genomic analysis of LUAD using TCGA data has revealed several important findings. The mutation frequencies observed align with previously reported rates, with TP53 alterations being the most common (50%), followed by KRAS (27%) and EGFR (13%) (Kadara et al., 2017). The significant association between TP53 mutations and poor survival is consistent with its known role as a tumor suppressor, suggesting that loss of TP53 function contributes to more aggressive disease behavior (Ma et al., 2016).

Pathologic stage emerged as the strongest predictor of survival, highlighting the continued importance of traditional staging in prognosis assessment. The integration of molecular features with clinical factors could enhance prognostic stratification, as demonstrated by the combined impact of stage and TP53 mutation status on survival outcomes (Chen et al., 2014).

Gene expression analysis revealed distinct patterns that may represent different molecular subtypes of LUAD. Further investigation of these expression signatures could provide insights into disease biology and potentially identify new therapeutic targets (Sholl et al., 2015).

This study's limitations include its retrospective nature, lack of treatment data. Nevertheless, the findings contribute to our understanding of the genomic landscape of LUAD and identify potential prognostic biomarkers that warrant further investigation in prospective studies (Govindan et al., 2012).

Conclusion

This comprehensive genomic characterization of lung adenocarcinoma confirms the importance of TP53 mutations in disease pathogenesis and prognosis. Pathologic stage remains the strongest predictor of survival, but molecular features can provide additional prognostic information. The integration of clinical factors with genomic alterations may improve patient stratification and guide therapeutic decisions in the era of precision medicine (Hirsch et al., 2017).

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